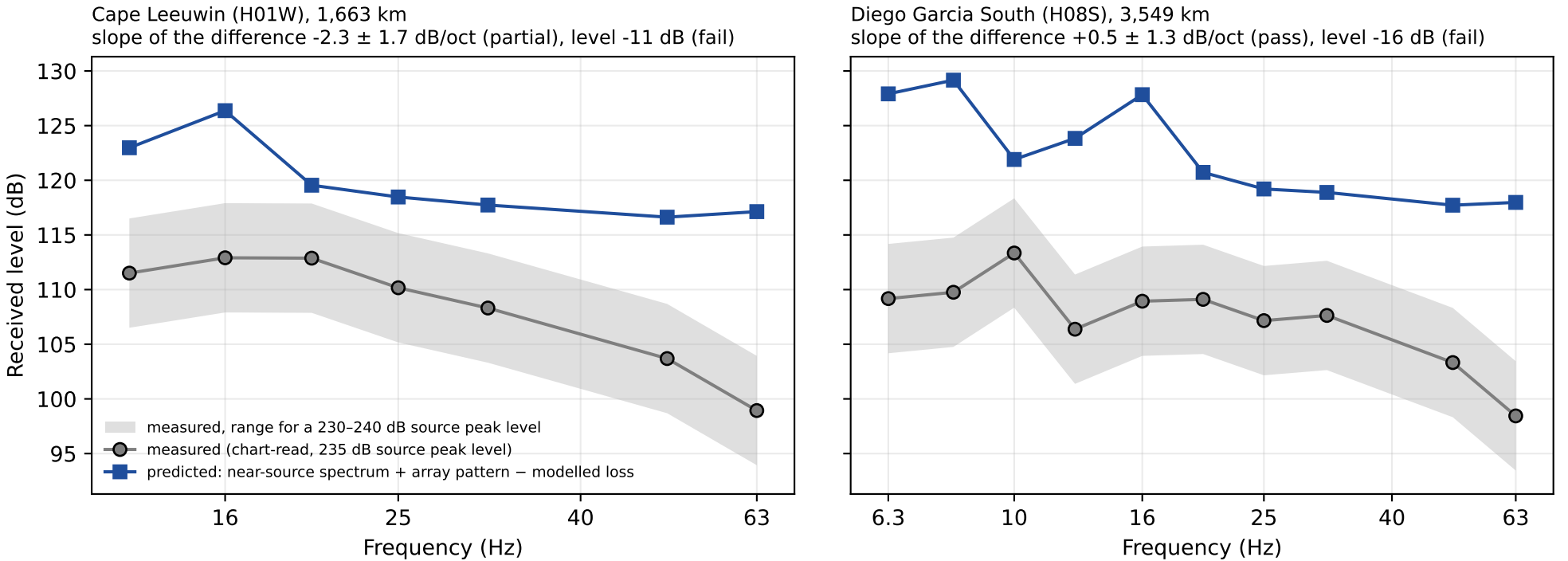
Airgun shots of October 2001 (site air9): predicted against measured received level at the two IMS stations, with the source spectrum taken from the ship's own near-source recording



What the chart shows: for one set of airgun shots fired in 2001, the sound level we predict at two listening stations (blue) against the level that was measured (grey). The source level now comes from a recording made next to the airguns on the ship. The slope of the two curves now agrees (within 3 dB per octave); the vertical offset does not, but the measured level is a peak level and the predicted level is a spectrum level, so the offset cannot be read as an error. Verdict under the rule fixed before the calculation: NOT EXPLAINED (the level test fails). Caveats C1 and C2 apply to every number.

Technical: RL_pred = S0 + AF − TL; S0 = EW0112 Fig. 2 Ball spectrum (DIGITISED, two shots, dB mean) + 20 log10(25.75 m) − near-field reference factor (OFR 95-228 20-gun layout, V^(1/3) weights, Ball 30 m ahead, 15 m deep, source 10 m); AF = horizontal array factor toward the station; TL = KRAKEN adiabatic incoherent modes, hard bottom, Francois-Garrison, shared GEBCO_2026/WOA23 (Oct) sections, receiver at triad depth. RL_obs = SL_B − TL_obs(Fig. 23, DIGITISED ±2 dB), SL_B = 235 (230–240 band). Bands: Fig. 23 span ∩ Fig. 2 readable (40 Hz excluded). Grades: audit criteria.md 940f1a3. Pre-registration e949144 (tl_recal_block9.py). Verdict NOT EXPLAINED; slope grid fraction 0.78 of 45. No MH370 impact posterior enters this chart (calibration event only; run J is the impact set the module predicts for). C1: propagation slope misfit (+10.4/+7.4 dB/oct uncorrected); C2: coupling η not calibrated.